

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times bool \times bool \times int \Rightarrow State \\ \text{c00} : Loc \\ \text{c10} : Loc \\ \text{c20} : Loc \\ \text{crit0} : Loc \\ \text{c01} : Loc \\ \text{c11} : Loc \\ \text{c21} : Loc \\ \text{crit1} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{ll}
 1 : & \text{state}(c00, p1, b0, b1, k) \rightarrow \text{state}(c10, p1, b01, b1, k) \quad [\neg(b01)] \\
 2 : & \text{state}(c10, p1, b0, b1, k) \rightarrow \text{state}(c20, p1, b0, b1, k) \quad [k \neq 0] \\
 3 : & \text{state}(c10, p1, b0, b1, k) \rightarrow \text{state}(\text{crit}0, p1, b0, b1, k) \quad [k = 0] \\
 4 : & \text{state}(c20, p1, b0, b1, k) \rightarrow \text{state}(c10, p1, b0, b1, k1) \quad [b1 \wedge k1 = 0] \\
 5 : & \text{state}(\text{crit}0, p1, b0, b1, k) \rightarrow \text{state}(c00, p1, b01, b1, k) \quad [b01] \\
 6 : & \text{state}(p0, c01, b0, b1, k) \rightarrow \text{state}(p0, c11, b0, b11, k) \quad [\neg(b11)] \\
 7 : & \text{state}(p0, c11, b0, b1, k) \rightarrow \text{state}(p0, c21, b0, b1, k) \quad [k \neq 1] \\
 8 : & \text{state}(p0, c11, b0, b1, k) \rightarrow \text{state}(p0, \text{crit}1, b0, b1, k) \quad [k = 1] \\
 9 : & \text{state}(p0, c21, b0, b1, k) \rightarrow \text{state}(p0, c11, b0, b1, k1) \quad [b0 \wedge k1 = 1] \\
 10 : & \text{state}(p0, \text{crit}1, b0, b1, k) \rightarrow \text{state}(p0, c01, b0, b11, k) \quad [b11] \\
 11 : & \text{state}(p0, p1, b0, b1, k) \rightarrow \text{success} \\
 12 : & \text{state}(\text{crit}0, \text{crit}1, b0, b1, k) \rightarrow \text{error}
 \end{array} \right.$$

$$\{13: \text{state}(c00, c01, \text{true}, \text{true}, k) \Rightarrow \text{success}[k = 0 \vee k = 1]\}$$

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times bool \times bool \times bitvec_1 \Rightarrow State \\ \text{c00} : Loc \\ \text{c10} : Loc \\ \text{c20} : Loc \\ \text{crit0} : Loc \\ \text{c01} : Loc \\ \text{c11} : Loc \\ \text{c21} : Loc \\ \text{crit1} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{ll}
 1 : & \text{state}(c00, p1, b0, b1, k) \rightarrow \text{state}(c10, p1, b01, b1, k) \quad [\neg(b01)] \\
 2 : & \text{state}(c10, p1, b0, b1, k) \rightarrow \text{state}(c20, p1, b0, b1, k) \quad [k \neq \#b0] \\
 3 : & \text{state}(c10, p1, b0, b1, k) \rightarrow \text{state}(\text{crit}0, p1, b0, b1, k) \quad [k = \#b0] \\
 4 : & \text{state}(c20, p1, b0, b1, k) \rightarrow \text{state}(c10, p1, b0, b1, k1) \quad [b1 \wedge k1 = \#b0] \\
 5 : & \text{state}(\text{crit}0, p1, b0, b1, k) \rightarrow \text{state}(c00, p1, b01, b1, k) \quad [b01] \\
 6 : & \text{state}(p0, c01, b0, b1, k) \rightarrow \text{state}(p0, c11, b0, b11, k) \quad [\neg(b11)] \\
 7 : & \text{state}(p0, c11, b0, b1, k) \rightarrow \text{state}(p0, c21, b0, b1, k) \quad [k \neq \#b1] \\
 8 : & \text{state}(p0, c11, b0, b1, k) \rightarrow \text{state}(p0, \text{crit}1, b0, b1, k) \quad [k = \#b1] \\
 9 : & \text{state}(p0, c21, b0, b1, k) \rightarrow \text{state}(p0, c11, b0, b1, k1) \quad [b0 \wedge k1 = \#b1] \\
 10 : & \text{state}(p0, \text{crit}1, b0, b1, k) \rightarrow \text{state}(p0, c01, b0, b11, k) \quad [b11] \\
 11 : & \text{state}(p0, p1, b0, b1, k) \rightarrow \text{success} \\
 12 : & \text{state}(\text{crit}0, \text{crit}1, b0, b1, k) \rightarrow \text{error}
 \end{array} \right\}$$

$$\{13: \text{state}(c00, c01, \text{true}, \text{true}, k) \Rightarrow \text{success} \mid [k = \#b0 \vee k = \#b1]\}$$